SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA0718**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer**.** Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT)**.** Discuss how ICMP assists in network troubleshooting and connectivity verification.

Aim

To develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer, configure the required IP addresses, and verify connectivity between the PCs using the ping command.

Software Required

• Cisco Packet Tracer

• A computer/laptop running Windows, Linux, or macOS

Network Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
Router0	FastEthernet0/0	192.168.1.1	255.255.255.0
Router0	FastEthernet1/0	192.168.2.1	255.255.255.0
PC1	FastEthernet0	192.168.2.10	255.255.255.0

Procedure

1. Open Cisco Packet Tracer and create a new workspace.
2. Place two PCs (PC0 and PC1) and one router (Router0) in the logical workspace.
3. Connect PC0 FastEthernet0 to Router0 FastEthernet0/0 and PC1 FastEthernet0 to Router0 FastEthernet1/0 using suitable Ethernet cables.
4. Configure PC0 by selecting Desktop → IP Configuration. Set IP address to 192.168.1.10, subnet mask to 255.255.255.0, and default gateway to 192.168.1.1.
5. Configure PC1 by selecting Desktop → IP Configuration. Set IP address to 192.168.2.10, subnet mask to 255.255.255.0, and default gateway to 192.168.2.1.
6. Open Router0 → CLI. Enter the following commands to configure the first interface:

enable

configure terminal

interface fastEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

7. Configure the second router interface using the following commands:

interface fastEthernet 1/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

8. Verify the router interfaces using the command:

show ip interface brief

Both FastEthernet0/0 and FastEthernet1/0 should show an operational status of up/up.

9. On PC0, open Desktop → Command Prompt and test connectivity to the router:

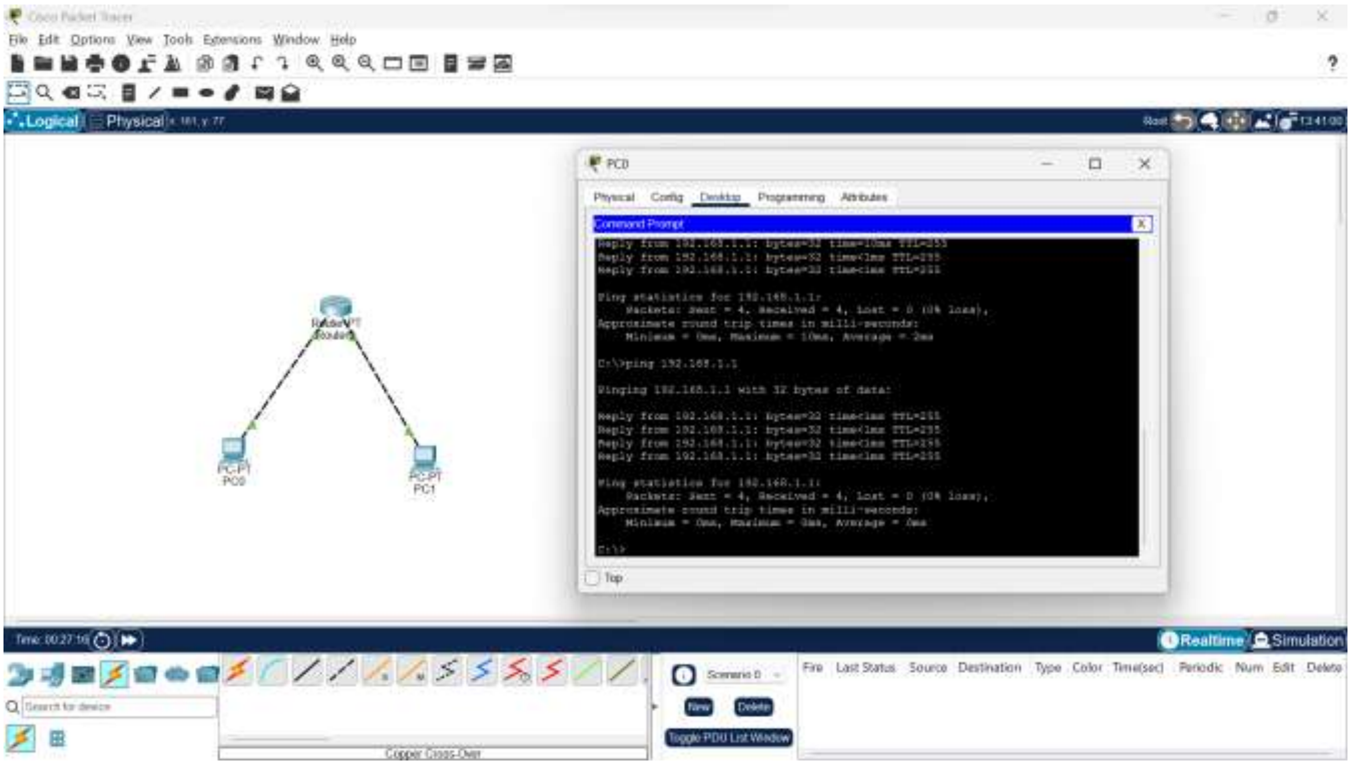
ping 192.168.1.1

10. From PC0, test connectivity to PC1:

ping 192.168.2.10

Successful replies confirm that PC0 can communicate with PC1 through the router.

Output



Result

The network consisting of two PCs connected through a router was successfully created and configured in Cisco Packet Tracer. IP addresses were assigned to the PCs and router interfaces, and connectivity between PC0 and PC1 was successfully verified using the ping command.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation